CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 7

Contents

1. Multivariate Linear Regression Model (Continued)
   - Multivariate Central Limit Theorem (CLT)
   - Asymptotic Normality of OLS Estimator in Multivariate Regression
     - Large Sample Distribution of $\hat{\boldsymbol{\beta}}$
     - Heteroskedasticity-robust Standard Errors
     - Confidence Intervals for Predicted Effects
     - Joint Hypotheses

2. Nonlinear Regression Functions
   - Regression Functions Linear in Parameters
     - Assumptions
     - Examples: Quadratic Regression, Polynomials, Logarithms
     - Interactions

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Multivariate Central Limit Theorem (CLT)

Thm. (Multivariate CLT)
Let $W_1, W_2, W_3, \ldots$ be a sequence of independent identically distributed $m \times 1$ random matrices (vectors) with

$$E(W_k) = \mu \;\text{ and }\; \mathrm{Var}(W_k) = \Sigma \;\text{ for } k = 1, 2, 3, \ldots,$$

where Σ is finite and positive definite.

Let

$$\bar{W}_n \doteq \frac{1}{n} \sum_{k=1}^{n} W_k$$

then

$$\sqrt{n}(\bar{W}_n - \mu) \xrightarrow{D} N(0, \Sigma).$$

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Asymptotic Normality of OLS Estimator in Multivariate Regression

Thm.
Given Multivariate Linear Regression Econometric model,

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{u} \quad \text{with no } \textit{multicollinearity}\text{, where}$$

$$\boldsymbol{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \boldsymbol{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & & & \ddots & \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}, \quad \text{and } \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix},$$

as $n \to \infty$, we have:

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{D} N(0, \Sigma_{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}), \quad \text{where } \Sigma_{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})} = \boldsymbol{Q}_X^{-1} \Sigma_V \boldsymbol{Q}_X^{-1},$$

where $\boldsymbol{Q}_X = \mathrm{E}[X_i X_i^{\mathsf{T}}]$ and $\Sigma_V = \mathrm{E}[V_i V_i^{\mathsf{T}}]$ for $V_i \doteq X_i u_i$.
Here, $X_i = (1 \ X_{i1} \ \dots \ X_{ik})^{\mathsf{T}}$ is the i^{th} observation.

Heteroskedasticity-robust Standard Errors

In order to get $\text{SE}(\hat{\beta}_j)$, we need to estimate diagonal elements of $\Sigma_{\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}$ via estimating matrices $\boldsymbol{Q}_X$ and Σ_V:

$$\hat{\Sigma}_{\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})} = \left(\frac{\boldsymbol{X}^\top \boldsymbol{X}}{n}\right)^{-1} \hat{\Sigma}_V \left(\frac{\boldsymbol{X}^\top \boldsymbol{X}}{n}\right)^{-1},$$

where

$$\hat{\Sigma}_V = \frac{1}{n-k-1} \sum_{i=1}^{n} X_i X_i^\top \hat{u}_i^2.$$

Then

$$\text{SE}(\hat{\beta}_j) = \sqrt{\left(\hat{\Sigma}_{\hat{\boldsymbol{\beta}}}\right)_{jj}}, \quad \text{where } \hat{\Sigma}_{\hat{\boldsymbol{\beta}}} = n^{-1} \hat{\Sigma}_{\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}.$$

Confidence Intervals for Predicted Effects

Let $\boldsymbol{d}$ be $(k+1) \times 1$ matrix. Assume we want to obtain a confidence interval for $\boldsymbol{d}^\mathsf{T}\boldsymbol{\beta}$, then

$$\sqrt{n}(\boldsymbol{d}^\mathsf{T}\hat{\boldsymbol{\beta}} - \boldsymbol{d}^\mathsf{T}\boldsymbol{\beta}) = \boldsymbol{d}^\mathsf{T}\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{D} N(0, \boldsymbol{d}^\mathsf{T}\Sigma_{\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}\boldsymbol{d})$$

and therefore

$$\boldsymbol{d}^\mathsf{T}\boldsymbol{\beta} = \boldsymbol{d}^\mathsf{T}\hat{\boldsymbol{\beta}} \pm z_{\alpha/2}\sqrt{\boldsymbol{d}^\mathsf{T}\hat{\Sigma}_{\hat{\boldsymbol{\beta}}}\boldsymbol{d}} \text{ with } (1-\alpha)100\% \text{ confidence.}$$

Joint Hypotheses

Let $\boldsymbol{R}$ be $q \times (k+1)$ matrix. Assume we want to test

$$H_0 : \boldsymbol{R\beta} = \boldsymbol{r} \;\; \text{vs.} \;\; H_1 : \boldsymbol{R\beta} \neq \boldsymbol{r}$$

for some $q \times 1$ vector $\boldsymbol{r}$, then one can use the following TS:

$$\mathsf{TS} = (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{r})^{\mathsf{T}} [\boldsymbol{R}\hat{\Sigma}_{\hat{\boldsymbol{\beta}}}\boldsymbol{R}^{\mathsf{T}}]^{-1} (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{r})/q,$$

which approximately follows F-distribution for large n:

$$\mathsf{TS} \xrightarrow{D} F_{q,\infty}.$$

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Nonlinear Regression Functions: Linear in Parameters

Assumptions:

We assume that $X_{i1}, X_{i2}, \dots, X_{ik}$ and Y_i are random and there is relation

$$Y_i = f(X_{i1}, X_{i2}, \dots, X_{ik}) + u_i,$$

where

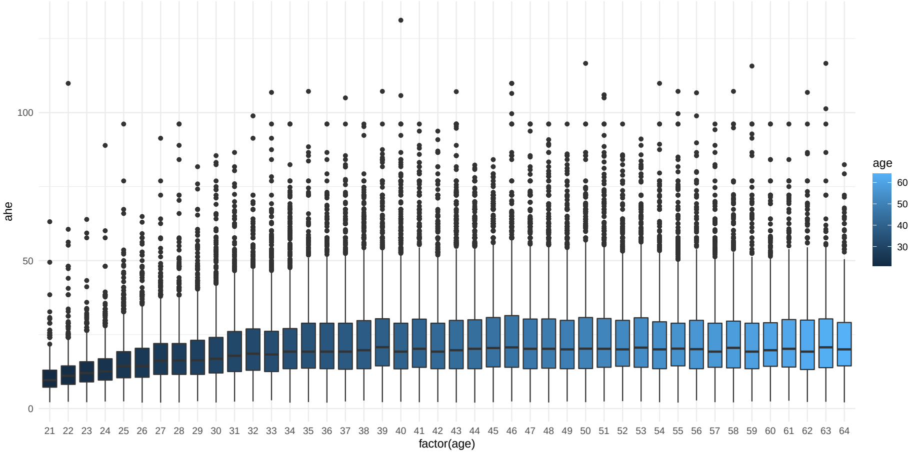
1. f is linear in coefficients, i.e.

$$f(X_{i1}, X_{i2}, \dots, X_{ik}) = \beta_0 + \beta_1 \underbrace{f_1(X_{i1}, \dots, X_{ik})}_{\text{1st regressor}} + \dots + \beta_K \underbrace{f_K(X_{i1}, \dots, X_{ik})}_{\text{Kth regressor}},$$

   where $f_1, \dots, f_K$ are independent of $\beta_0, \beta_1, \dots, \beta_K$

2. *exogeneity* assumption: $\mathrm{E}[u_i | f_1(X_{i1}, \dots, X_{ik}), \dots, f_K(X_{i1}, \dots, X_{ik})] = 0$
3. $(X_{i1}, X_{i2}, \dots, X_{ik}, Y_i)$ is i.i.d. sample drawn from a joint distribution
4. $\mathrm{E}[f_1^4(X_{i1}, \dots, X_{ik})], \dots, \mathrm{E}[f_K^4(X_{i1}, \dots, X_{ik})]$ and $\mathrm{E}[Y_i^4]$ are nonzero and finite
5. there is no *multicollinearity* among $1, f_1(X_{i1}, \dots, X_{ik}), \dots, f_K(X_{i1}, \dots, X_{ik})$

Example 1: Quadratic Regression



Example 1: Quadratic Regression

$$\text{ahe}_i = \beta_0 + \beta_1 \underbrace{\text{age}_i}_{f_1(\text{age}_i)} + \beta_2 \underbrace{\text{age}_i^2}_{f_2(\text{age}_i)} + u_i$$

```
Call:
lm(formula = ahe ~ age + I(age^2), data = df)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.073e+01  7.406e-01  -14.48   <2e-16 ***
age          1.431e+00  3.674e-02   38.95   <2e-16 ***
I(age^2)    -1.449e-02  4.351e-04  -33.31   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.28 on 63192 degrees of freedom
Multiple R-squared:  0.04724, Adjusted R-squared:  0.04721
F-statistic:  1566 on 2 and 63192 DF,  p-value: < 2.2e-16
```

Examples: Quadratic Regression, Polynomials, Logarithms

Example 1: Quadratic Regression

$$\text{ahe}_i = \beta_0 + \beta_1 \underbrace{\text{age}_i}_{f_1(\text{age}_i)} + \beta_2 \underbrace{\text{age}_i^2}_{f_2(\text{age}_i)} + u_i$$

According to this model, if $\text{age}_i = a$ and $\Delta\text{age}_i = 1$, the expected change in average hourly earnings is

$$\begin{aligned}
\Delta\widehat{\text{ahe}}_i = & \left(\hat{\beta}_0 + \hat{\beta}_1 \,\text{age}_i + \hat{\beta}_2 \,\text{age}_i^2\right)\Bigg|_{\text{age}_i = a+1} - \left(\hat{\beta}_0 + \hat{\beta}_1 \,\text{age}_i + \hat{\beta}_2 \,\text{age}_i^2\right)\Bigg|_{\text{age}_i = a} \\
= & \hat{\beta}_1 \; ((a+1) - a) + \hat{\beta}_2 \; \left((a+1)^2 - a^2\right) \\
= & \hat{\beta}_1 + \hat{\beta}_2 \,(2a+1)
\end{aligned}$$

Examples: Quadratic Regression, Polynomials, Logarithms

Example 1: Quadratic Regression

How can one obtain the standard error of the estimated change

$$\Delta\widehat{\mathsf{ahe}}_i = \hat{\beta}_1 + \hat{\beta}_2\,(2a+1)?$$

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$$\Delta \widehat{\mathsf{ahe}}_i = \hat{\beta}_1 + \hat{\beta}_2\,(2a+1)?$$

- Method 1: We can re-write the model as follows:

$$\begin{aligned} \mathsf{ahe}_i =& \beta_0 + \beta_1\,\mathsf{age}_i + \beta_2\,\mathsf{age}_i^2 + u_i \\ =& \beta_0 + \underbrace{(\beta_1 + \beta_2\,(2a+1))}_{\text{slope of interest}}\mathsf{age}_i + \beta_2\left(\mathsf{age}_i^2 - (2a+1)\mathsf{age}_i\right) + u_i \end{aligned}$$

and re-run the regression model on age_i and $\left(\mathsf{age}_i^2 - (2a+1)\mathsf{age}_i\right)$ regressors to get $\mathsf{SE}(\Delta\widehat{\mathsf{ahe}}_i)$.

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  and re-run the regression model on age_i and $\left(\text{age}_i^2 - (2a+1)\text{age}_i\right)$ regressors to get $\text{SE}(\Delta\widehat{\text{ahe}}_i)$.

- Method 2: Based on F-statistic for $H_0 : \beta_1 + \beta_2\,(2a+1) = 0$ vs. $H_1 : \beta_1 + \beta_2\,(2a+1) \neq 0$ we get $\text{SE}(\Delta\widehat{\text{ahe}}_i) = |\Delta\widehat{\text{ahe}}_i|/\sqrt{F}$.
  Note:
  $F = t^2 = \left[\left(\hat{\beta}_1 + \hat{\beta}_2\,(2a+1)\right)/\text{SE}(\hat{\beta}_1 + \hat{\beta}_2\,(2a+1))\right]^2 = \left[\Delta\widehat{\text{ahe}}_i/\text{SE}(\Delta\widehat{\text{ahe}}_i)\right]^2.$

Examples: Quadratic Regression, Polynomials, Logarithms

Example 2: Polynomial Regression

$$\text{ahe}_i = \beta_0 + \beta_1 \underbrace{\text{age}_i}_{f_1(\text{age}_i)} + \beta_2 \underbrace{\text{age}_i^2}_{f_2(\text{age}_i)} + \beta_3 \underbrace{\text{age}_i^3}_{f_3(\text{age}_i)} + \beta_4 \underbrace{\text{age}_i^4}_{f_4(\text{age}_i)} + u_i$$

```
Call:
lm(formula = ahe ~ age + I(age^2) + I(age^3) + I(age^4), data = df)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -4.732e+01  9.353e+00  -5.059 4.22e-07 ***
age          4.363e+00  9.752e-01   4.474 7.69e-06 ***
I(age^2)    -9.001e-02  3.680e-02  -2.446   0.0145 *
I(age^3)     6.607e-04  5.974e-04   1.106   0.2688
I(age^4)    -7.285e-07  3.530e-06  -0.206   0.8365
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.26 on 63190 degrees of freedom
Multiple R-squared:  0.05006, Adjusted R-squared:   0.05
F-statistic: 832.5 on 4 and 63190 DF,  p-value: < 2.2e-16
```

Examples: Quadratic Regression, Polynomials, Logarithms

Example 2: Polynomial Regression

```
fit1 <- lm(ahe ~ age, data = df)
fit2 <- lm(ahe ~ age + I(age^2), data = df)
fit3 <- lm(ahe ~ age + I(age^2) + I(age^3), data = df)
fit4 <- lm(ahe ~ age + I(age^2) + I(age^3) + I(age^4), data = df)

library(stargazer)
stargazer(fit1, fit2, fit4, type = "text")
```

Example 2: Polynomial Regression

	Dependent variable:		
	ahe		
	(1)	(2)	(3)
age	0.218***	1.431***	4.363***
	(0.005)	(0.037)	(0.975)
I(age2)		-0.014***	-0.090**
		(0.0004)	(0.037)
I(age3)			0.001
			(0.001)
I(age4)			-0.00000
			(0.00000)
Constant	12.934***	-10.725***	-47.321***
	(0.211)	(0.741)	(9.353)
Observations	63,195	63,195	63,195
R2	0.031	0.047	0.050
Adjusted R2	0.030	0.047	0.050
Residual Std. Error	13.393 (df = 63193)	13.277 (df = 63192)	13.257 (df = 63190)
F Statistic	1,988.816*** (df = 1; 63193)	1,566.481*** (df = 2; 63192)	832.454*** (df = 4; 63190)
Note:			*p<0.1; **p<0.05; ***p<0.01

Examples: Quadratic Regression, Polynomials, Logarithms

Example 3: Logarithms

$$\text{ahe}_i = \beta_0 + \beta_1 \underbrace{\ln(\text{age}_i)}_{f_1(\text{age}_i)} + u_i$$

```
Call:
lm(formula = ahe ~ log(age), data = df)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.8808     0.7115  -18.11   <2e-16 ***
log(age)      9.4480     0.1919   49.23   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.35 on 63193 degrees of freedom
Multiple R-squared:  0.03693, Adjusted R-squared:  0.03692
F-statistic:  2423 on 1 and 63193 DF,  p-value: < 2.2e-16
```

Example 4: Nonlinear Controls

```
fit1 <- lm(ahe ~ yrseduc, data = df)
fit2 <- lm(ahe ~ yrseduc + female + northeast + midwest + south + age, data = df)
fit3 <- lm(ahe ~ yrseduc + female + northeast + midwest + south + age + I(age^2), data = df)

library(stargazer)
stargazer(fit1, fit2, fit3, type = "text")
```

	Dependent variable:		
	ahe		
	(1)	(2)	(3)
yrseduc	2.383***	2.392***	2.364***
	(0.019)	(0.018)	(0.018)
female		-5.710***	-5.685***
		(0.093)	(0.093)
northeast		0.420***	0.347**
		(0.139)	(0.138)
midwest		-1.917***	-1.987***
		(0.133)	(0.132)
south		-1.346***	-1.375***
		(0.124)	(0.123)
age		0.192***	1.192***
		(0.004)	(0.032)
I(age2)			-0.012***
			(0.0004)
Constant	-11.083***	-15.906***	-35.004***
	(0.270)	(0.315)	(0.682)
Observations	63,195	63,195	63,195
R2	0.197	0.268	0.280
Adjusted R2	0.197	0.268	0.280
Residual Std. Error	12.188 (df = 63193)	11.634 (df = 63188)	11.544 (df = 63187)
F Statistic	15,510.130*** (df = 1; 63193)	3,864.908*** (df = 6; 63188)	3,506.820*** (df = 7; 63187)

Interactions

Interactions between independent variables:

- Two binary variables:

$$Y_i = \beta_0 + \beta_1 D_{i1} + \beta_2 D_{i2} + \beta_3 \underbrace{(D_{i1} \times D_{i2})}_{\text{interaction term}} + u_i$$

- Continuous and binary variables:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 \underbrace{(X_i \times D_i)}_{\text{interaction term}} + u_i$$

- Two continuous variables:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 \underbrace{(X_{i1} \times X_{i2})}_{\text{interaction term}} + u_i$$